



Extracting inflation expectations and inflation risk premia from the term structure: A joint model of the UK nominal and real yield curves

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ABSTRACT

This paper analyses the UK interest rate term structure over the period since October 1992, when the United Kingdom adopted an explicit inflation target, using an affine term structure model estimated using both government bond yields and survey data. The model imposes no-arbitrage restrictions across nominal and real yields, which enables interest rates to be decomposed into expected real policy rates, expected inflation, real term premia and inflation risk premia. The model is used to shed light on major developments over the period, including the impact of Bank of England independence and the low real bond yield 'conundrum'.

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1. Introduction

The nominal and real interest rate term structures implied by UK government conventional and index-linked bonds can potentially provide monetary policy makers¹ with a great deal of information about financial market expectations of both future interest rates and inflation. The nominal and real term structures embody market expectations of future nominal and real interest rates respectively, while the difference between the two – the inflation term structure – embodies information about inflation expectations. Extracting this information, however, is complicated by the fact that the interest rate term structure may also reflect inflation risk premia and real term premia.²

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¹ Our focus is on monetary policy analysis, but of course information in the yield curve has a number of other uses; for example, it is important for risk management in financial institutions (see, e.g., Drehmann et al. (2009)).

² Government bond yields, particularly those for index-linked bonds, may also embody liquidity premia (the extra return investors require to compensate them for expected future illiquidity and illiquidity risk), but such premia are unlikely to have been a material influence on the UK term structure over most of our sample period and they are not explicitly incorporated into our modelling framework. We do, however, discuss the impact on our model estimates of specific episodes of extreme financial turbulence that may have affected liquidity premia, including the market disruptions that began in summer 2007 towards the end of our estimation period.

The main contribution of this paper is to estimate a joint model of the UK nominal and real term structures over the period that the United Kingdom has had an explicit inflation target, enabling us to decompose nominal forward rates into expected real policy (risk-free) rates, expected inflation, real term premia and inflation risk premia since October 1992. Although we are not the first to do so, there are surprisingly few previous papers that have estimated theoretically consistent term structure models using UK data on both index-linked and nominal bond yields. One earlier example is a paper by Gong et al. (1998), but the generalised CIR (Cox et al., 1985) model specification that they adopt is very restrictive and has been shown to fit term structure data poorly. Evans (2003) estimates an extended Vasicek (1977) model that incorporates Markov-switching regimes. But while that model allows term and inflation risk premia to vary over time according to three regimes, this set-up is still rather restrictive. An unpublished paper by Risa (2001), applies a more flexible essentially affine model, similar to our own, to modelling UK data from 1983 to 1999, but does not incorporate survey information on inflation expectations as we do (see discussion below). Moreover, given his sample period, Risa does not shed much light on the impact of Bank of England independence on the term structure of interest rates and does not analyse the reasons for the period of unusually low long-term real interest rates – christened the bond yield 'conundrum' by Greenspan (2005) – that began in 2004.

Our proposed model is based on the so-called essentially affine class of term structure models (see, e.g., Duffee (2002)). The approach has two main elements. First, we assume that UK nominal and real bond markets are arbitrage free, so that it is not possible to make risk-free profits from trading combinations of real and/or nominal bonds.³ Second, we assume that bonds are priced by a stochastic discount factor (SDF) that takes a particularly flexible form, where the market price of risk is a linear function of the observable and unobservable factors in the model. A consequence of the second assumption, and the main implication of the essentially affine model, is that in our model bond prices, and thus yields, are linearly related to inflation and a small set of unobservable latent factors. We favour latent factors rather than macro-factors, partly because this approach has been shown to provide a better statistical fit of term structure data and partly because by taking an agnostic approach to the underlying factors driving yields the resulting model may be less prone to misspecification.⁴

We assume that two latent factors drive movements in expected real short rates and that the same two factors and two additional ones (one retail prices index (RPI) inflation, the other unobservable) drive the nominal curve and real term premia. An important feature of the model is that the same real SDF is assumed to price both real and nominal bond yields. To the extent that institutional investors have preferred habitats for index-linked bonds and demand/supply imbalances push prices away from fundamentals, this assumption may not be an accurate description of the real world. However, if the importance of demand/supply imbalances changes over time and is not a permanent feature, the ability of the model fit to various segments of the forward curve may enable us to identify the emergence of such non-fundamental or market segment-specific factors.

Joyce et al. (2008) apply a similar essentially affine model to the UK real term structure in isolation, in order to investigate the emergence of unusually low long real interest rates during 2004–05. One disadvantage of applying the affine modelling framework purely to real yields is the lack of available shorter maturity index-linked bonds. In this paper, by including nominal bond yields and inflation in the model, we are able to derive model-based estimates of short-maturity real rates. Moreover, estimating a joint model provides a number of advantages because it ensures consistency between the two term structures by imposing no arbitrage across them. And by modelling the dynamics of inflation expectations as a function of the information that drives real interest rates, as well as nominal rates and inflation, we use current information in the whole term structure to extract measures of expected inflation and inflation premia.

To reduce the possibility of encountering instability in term structure behaviour resulting from changes in the United Kingdom's monetary framework, we limit our sample to the period since October 1992, during which the United Kingdom has operated an inflation target.⁵ However, this means there is potentially a small sample problem. As Kim and Orphanides (2005) demonstrate, small sample bias in term structure models can lead to implausible implications for the model-implied decompositions of forward rates into expected future short rates and term premia,

and in particular to low persistence in estimated expected short rates. In their application to the US yield curve, Kim and Orphanides (2005) advocate including survey data on the future path of interest rate expectations, as a way of supplementing the available time-series data on yields.⁶ In our paper we incorporate bi-annual Consensus survey information on expected average inflation five to 10 years ahead, as an additional information variable, which helps to identify long-run inflation expectations. The implicit assumption is that the long-run inflation expectations of bond market participants will be the same as those of the economic and financial forecasters surveyed by Consensus forecasts. But we include an error term to allow long-term inflation expectations from the model to differ from those of the survey, so that expectations need only be the same on average over the sample. So, although our model incorporates survey information, the model forecasts are not always in line with the surveys. And, while the long-term survey information is only available every six months, our model has the advantage that it provides monthly estimates of expected inflation at any horizon.

The paper is structured as follows: Section 2 sets out the theoretical relationships between nominal and real yields and the SDF that obtain under no arbitrage and shows how we can use them to decompose interest rates into expected future short rates and premia. We also discuss the real SDF imposed in the essentially affine term structure literature. In Section 3, we discuss the econometric methodology used to estimate the model and derive the link between bond yields and the set of variables explaining yields. In Section 4 we describe the data used for our empirical analysis. Section 5 presents the results and Section 6 concludes. Some of the main mathematical derivations are explained in the Appendix at the end of the paper.

2. Theory and the EATS model

In this section we establish the relationship between nominal and real bond prices and the real SDF in the absence of arbitrage possibilities. We show how we can decompose nominal and real interest rates into expected future short interest rates and term premia and then how the difference between nominal and real interest rates, i.e. inflation breakeven rates, can be broken down into inflation expectations, inflation risk premia and an inflation convexity effect. Finally, we describe the modelling approach adopted in this paper, which is based on an essentially affine term structure (EATS) model.

2.1. The link between the nominal and real SDFs under no arbitrage

In an arbitrage-free environment, where all risk-free profit opportunities are eliminated, we can think of investors as pricing assets according to the fundamental asset pricing equation, i.e. by the discounted present value of their future pay-offs (for discussion on this see, e.g., Cochrane (2005)). So the current price of a zero-coupon real bond, denoted by $P_t^{n,R}$, that pays one unit of the consumption good when it matures in period $t+n$ is given by

$$P_t^{n,R} = E_t[M_{t+1} \cdot M_{t+2} \cdots M_{t+n}], \quad (1)$$

where M_{t+j} denotes the real SDF in period j . This no-arbitrage condition corresponds to the Euler condition in a representative agent

³ As we show below, the assumption of no arbitrage enables us to decompose interest rates into expectations of future risk free rates and term premia. But, as well as being theoretically attractive, there is some evidence that imposing no-arbitrage restrictions significantly improves out-of-sample interest rate forecasts (see, e.g., Almeida and Vicente (2008)).

⁴ Most macro-factor models of the term structure find it necessary to include additional latent factors to fit longer-maturity yields (see, e.g., Ang and Piazzesi (2003)).

⁵ Using a long run of data covering the period from 1881 to 2003, Kanas (2008) finds that regime switching in the dynamics of US and UK real interest rates is driven by changes in inflation and political regimes.

⁶ In a subsequent paper, Kim and Wright (2005) incorporate survey information on both expected policy rates and inflation into their model of the US nominal and real term structures, but they do not incorporate market data on US index-linked bonds, so-called Treasury Inflation Protected Securities (TIPS), because of the lack of a long enough back run of data. More recently, D'Amico et al. (2008) have attempted to estimate a model using TIPS data. However, the sample available to them is still quite short (only 9 years) and TIPS appear to contain a sizable liquidity premium until quite recently.

macro-model, where the real SDF is often referred to as the inter-temporal marginal rate of substitution, as it indicates the investor's willingness to substitute consumption over time.

The price of a zero-coupon nominal bond, $P_t^{n,N}$, is also equal to the expectation of its discounted real pay-off, but we now also have to allow for changes in the price level:

$$P_t^{n,N} = E_t \left[\frac{M_{t+1} Q_t}{Q_{t+1}} \cdot \frac{M_{t+2} Q_{t+1}}{Q_{t+2}} \cdots \frac{M_{t+n} Q_{t+n-1}}{Q_{t+n}} \right], \quad (2)$$

where Q_t represents the general price level at time t . Nominal bond prices can be thought of as reflecting expectations of the nominal SDF, where the link between the nominal SDF, denoted with superscript *, and the real SDF is given by

$$M_{t+j}^* = M_{t+j} \frac{Q_{t+j-1}}{Q_{t+j}}. \quad (3)$$

2.2. Expectations, term premia and inflation risk premia

We now want to decompose interest rates into their expected future short rate component and real term premia/inflation premia. Taking logs of both sides of Eq. (1), we obtain the following relationship between the log price and the real SDF, which holds up to a second-order approximation for any distribution of the SDF:⁷

$$p_t^{n,R} = \ln E_t[M_{t+1} \cdot M_{t+2} \cdots M_{t+n}] = E_t \left[\sum_{j=1}^n m_{t+j} \right] + \frac{1}{2} V_t \left[\sum_{j=1}^n m_{t+j} \right], \quad (4)$$

where $p_t^{n,R} = \ln(P_t^{n,R})$ and $m_{t+j} = \ln(M_{t+j})$. Using the relationship between yields and prices $y_t^{n,R} = -\frac{p_t^{n,R}}{n}$, where $y_t^{n,R}$ is the current real yield on a bond maturing at n , it turns out that the real one-period short rate is given by $y_{t+j}^{1,R} = -E_{t+j}(m_{t+j+1}) - \frac{1}{2} V_{t+j}(m_{t+j+1})$. And the relationship between the n -period real yield, expected future real short rates and the real term premium is given by

$$y_t^{n,R} = \frac{1}{n} \left(E_t \left[\sum_{j=1}^n y_{t+j-1}^{1,R} \right] - \sum_{j=2}^n \text{Cov}_t \left[\sum_{k=1}^{j-1} m_{t+k}, m_{t+j} \right] \right). \quad (5)$$

The real yield reflects the average of expected future one-period real interest rates and the average real forward term premium over the life of the bond. A similar expression can be shown to apply for the nominal yield, $y_t^{n,N}$:

$$y_t^{n,N} = \frac{1}{n} \left(-E_t \left[\sum_{j=1}^n (m_{t+j} - \pi_{t+j}) \right] - \frac{1}{2} V_t \left[\sum_{j=1}^n (m_{t+j} - \pi_{t+j}) \right] \right), \quad (6)$$

where $\pi_{t+j} = \ln(\frac{Q_{t+j}}{Q_{t+j-1}})$ is log inflation. Subtracting (5) from (6) and rearranging gives the following expression for the inflation breakeven rate:

$$y_t^{n,N} - y_t^{n,R} = \frac{1}{n} \left(E_t \left[\sum_{j=1}^n \pi_{t+j} \right] - \frac{1}{2} V_t \left[\sum_{j=1}^n \pi_{t+j} \right] + \text{Cov}_t \left[\sum_{j=1}^n m_{t+j}, \sum_{j=1}^n \pi_{t+j} \right] \right). \quad (7)$$

So the breakeven rate is composed of three terms: the average expected log inflation rate over the life of the bond, the average inflation convexity effect and the average inflation risk premium. The Fisher hypothesis ignores the last two terms. But if the inflation convexity effect or the inflation risk premium are different from zero, it is clear that we cannot get a direct reading of inflation expectations from breakeven rates.

For many purposes, it is more useful to focus on forward interest rates, the rates implied for future time periods, rather than spot rates, which refer to average rates over a period. We can derive similar decompositions for forward rates because the difference between the n -period nominal and real spot rates is simply the average of the difference between the n one-period forward rates between t and $t+n$. Hence

$$y_t^{n,N} - y_t^{n,R} = \frac{1}{n} \left(\sum_{j=1}^n (f_t^{j,N} - f_t^{j,R}) \right), \quad (8)$$

where $f_t^{j,N}$ denotes the implied one-period nominal rate, j periods ahead, and $f_t^{j,R}$ is the equivalent real rate.

2.3. The real SDF in an essentially affine term structure model

If we knew how the SDF was linked to marginal utility and the dynamics of macroeconomic variables that affect marginal utility, it would be straightforward to decompose the term structure into expected future short rates and premia. In practice, however, representative agent models with conventional preferences have had limited success in characterising the dynamics of asset price movements (see Rudebusch and Swanson (2008) for an application to the US term structure).

An alternative approach assumes that there exists at least one real SDF that prices all bonds. But rather than specifying the underlying utility function, it is assumed that the dynamics of the real SDF can be summarised by a number of observable macroeconomic and/or latent factors. The affine term structure literature assumes that this real SDF, which satisfies pricing Eqs. (1) and (2),⁸ takes the form:

$$\ln M_{t+1} = m_{t+1} = -(\bar{r} + \gamma'(z_t - \mu)) - \frac{A_t' \Omega A_t}{2} - A_t' \Omega_t^{1/2} \varepsilon_{t+1} \quad (9)$$

so that the log of the nominal SDF is given by

$$m_{t+1}^* = m_{t+1} - \pi_{t+1} = -(\bar{r} + \gamma'(z_t - \mu)) - \frac{A_t' \Omega A_t}{2} - A_t' \Omega_t^{1/2} \varepsilon_{t+1} - \pi_{t+1}, \quad (10)$$

where $\bar{r}$ is the long-run level of the real short rate, $(z_t - \mu)$ is an $(N \times 1)$ vector of observable and/or unobservable (latent) variables with a zero mean, ε_{t+1} is an $(N \times 1)$ vector of shocks to the observable and/or unobservable (latent) variables and Ω_t is the conditional covariance matrix of these shocks, which may vary over time. The market price of risk is represented by $A_t' \Omega_t^{1/2}$. One price of risk specification, the essentially affine term structure (EATS) model proposed by Duffee (2002), assumes that A_t is linear in the factors. If, for example, we had two factors then A_t would be given by

$$A_t = \lambda + \beta(z_t - \mu) = \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} + \begin{bmatrix} \beta_{11} & \beta_{12} \\ \beta_{21} & \beta_{22} \end{bmatrix} \begin{bmatrix} z_{1,t} - \mu_1 \\ z_{2,t} - \mu_2 \end{bmatrix}. \quad (11)$$

The A_t vector is crucial for determining the time-variation in risk premia in an essentially affine model, when the covariance matrix between the shocks, Ω , is constant. If Ω is constant and $\beta_{11} = \beta_{12} = \beta_{21} = \beta_{22} = 0$, the conditional variance of the real SDF is constant and bond risk premia, at all maturities, are constant. If, in addition, $\lambda_1 = \lambda_2 = 0$, then the conditional variance of the real SDF is zero (i.e. the real SDF is non-stochastic) and risk premia are zero.

The dynamics of the z_t variables determine the dynamics of the real SDF, but not all of the variables need explain the expected real SDF. So γ' is a $(1 \times N)$ vector with ones in the rows corresponding

⁷ The approximation is exact in the case where the real SDF is conditionally log-normal.

⁸ See Duffee and Kan (1996) for a more thorough discussion of affine term structure models.

to the variables assumed to drive the expected real SDF and zeros elsewhere. The form of the SDF in this case is considerably more general than the SDF in a representative agent model, but it can be given a similar interpretation in terms of marginal utility.

Given this specification of the real SDF, the key feature of an EATS model is that expected future short interest rates, real term premia and inflation risk premia are all linear (affine) in the level of the N unobservable and/or observable (latent) variables. Real yields are given by

$$y_t^{n,R} = \bar{A}_n + \bar{B}_n'(z_t - \mu) \quad (12)$$

and nominal bond yields are given by

$$y_t^{n,N} = \bar{A}_n^* + \bar{B}_n^{*'}(z_t - \mu). \quad (13)$$

where $\bar{A}_n$ and $\bar{A}_n^*$ are scalars and $\bar{B}_n$ and $\bar{B}_n^*$ are $(N \times 1)$ parameter matrices that are obtained recursively, imposing no arbitrage across yields with different maturities. The recursive relationship between bonds with different maturities will depend on the underlying parameters in the SDF. The derivation of these equations is discussed in detail in Appendix.

3. Econometric modelling

To estimate a joint model of the nominal and real term structures, we need to specify the dynamics of the factors driving the real SDF and to make an assumption about how bond market investors form their inflation expectations. In this section we discuss these assumptions and then explain how we can estimate the resulting term structure model using the Kalman filter.

3.1. Modelling the factor dynamics

We want to be able to fit the nominal and real yield curves well, in order to be able to say something meaningful about the difference between the two – inflation expectations, inflation risk premia and the inflation convexity effect (though our estimates suggest this last element is small). For this reason, we use mainly latent variables rather than macro-variables, as the former are generally found to be necessary to fit the longer end of the yield curve closely. We assume that two latent factors drive movements in the expected real short rates and that the same two factors and two additional ones (one inflation, the other unobservable) drive the nominal curve and real term premia.⁹ The vector of state variables driving real and nominal yields is therefore given by

$$(z_t - \mu)' = [z_{1,t} - \mu_1 \quad z_{2,t} - \mu_2 \quad z_{3,t} - \mu_3 \quad z_{4,t} - \mu_4], \quad (14)$$

where $z_{1,t}$, $z_{2,t}$, and $z_{3,t}$ are latent factors with different time-series dynamics and the fourth factor, $z_{4,t}$, is inflation. As commonly assumed, we specify the dynamics of the factors as a first-order VAR with normally distributed errors:

$$(z_{t+1} - \mu) = \Phi(z_t - \mu) + \Omega^{1/2} \varepsilon_{t+1} \quad \varepsilon_{t+1} \sim NID(0, I_4), \quad (15)$$

where I_4 is a (4×4) identity matrix and the parameter matrices are given by

$$\Phi = \begin{bmatrix} \Phi_{11} & 0 & 0 & 0 \\ \Phi_{21} & \Phi_{22} & 0 & 0 \\ \Phi_{31} & \Phi_{32} & \Phi_{33} & 0 \\ \Phi_{41} & \Phi_{42} & \Phi_{43} & \Phi_{44} \end{bmatrix}, \quad \Omega = \begin{bmatrix} \sigma_1^2 & 0 & 0 & 0 \\ 0 & \sigma_2^2 & 0 & 0 \\ 0 & 0 & \sigma_3^2 & 0 \\ 0 & 0 & 0 & \sigma_4^2 \end{bmatrix},$$

$$\varepsilon_{t+1} = \begin{bmatrix} \varepsilon_{1,t+1} \\ \varepsilon_{2,t+1} \\ \varepsilon_{3,t+1} \\ \varepsilon_{4,t+1} \end{bmatrix}. \quad (16)$$

We assume that the variance of the factor shocks is constant. This is a common assumption in the term structure literature and seems reasonable given that we estimate the model over a relatively short sample period. Moreover, it eases estimation considerably. The additional assumptions that Ω is diagonal and that Φ is lower triangular are necessary for the factors to be identifiable given the use of latent variables. By allowing the off-diagonal elements of the Φ matrix to be non-zero, the factors are allowed to be correlated with each other. Notice that all three latent factors feed into the determination of inflation, which is the fourth factor.

3.2. Inflation expectations in the model

As is evident from Eq. (3), the relevant inflation variable for pricing bonds is the one-period log change in prices. As we will be working with monthly interest rate data, the relevant inflation measure to model is therefore month-on-month inflation. Inflation in our model is assumed to follow a vector auto regression (fourth row in Eq. (15)), subject to the restriction of no arbitrage, which ensures that inflation expectations at any future maturity have to be consistent with the dynamics of the factors that determine the nominal and real interest rates at that maturity. The advantage of modelling actual inflation, rather than treating inflation as a latent factor, is that inflation expectations and inflation risk premia are identified in a way consistent with observed inflation dynamics.

In the model we also include survey expectations of average RPI inflation 5–10 years ahead. The motivation for including survey expectations in term structure models of this nature is cogently explained in Kim and Orphanides (2005). The main problem is that we inevitably have to estimate the term structure model over a short sample, while interest rates themselves are highly persistent.¹⁰ This tends to lead to model estimates, which underestimate the persistence of expected short interest rates, and this in turn will have an impact on the decomposition of long-term interest rates into expectations and term premia.

We estimate our term structure model using monthly data from October 1992, when the UK inflation-targeting framework began, to February 2008. Since our sample period covers 15 years of monthly data, it is quite short and contains little business cycle variation. As a result, it may be difficult for any econometric model to attribute time-variation in long-term interest rates into expectations and premia. This is compounded by the fact that the EATS model implies a relationship between yields and factors that is highly non-linear in the underlying risk parameters, reflecting the restrictions that follow from the no-arbitrage condition. As a result, this class of term structure model is hard to estimate in short samples, as there are typically a number of different local maxima.

Incorporating survey information on long-horizon inflation expectations (from Consensus forecasts) into our model helps identify whether movements in breakevens are due to inflation

⁹ Most latent or macro-factor affine models of the nominal term structure use three factors (see, e.g., Lildholdt et al. (2007)), but since we are imposing restrictions across the real and nominal term structures we want to allow for the additional flexibility of an extra factor. In their study of the UK real term structure Joyce et al. (2008) find that two factors are sufficient to capture virtually all the dynamics of the real term structure. We allow shocks from each of the model's factors to affect the real pricing kernel, which introduces additional flexibility in fitting real yields.

¹⁰ For a discussion of the importance of accounting for time-variation in long-horizon forecasts of interest rates, see Kozicki and Tinsley (2001).

expectations or inflation premia.¹¹ Although we implicitly assume that bond market investors' expectations of average inflation 5–10 years ahead will be the same as those of the survey respondents, we include an error term to allow long-term inflation expectations from the model to differ from those of the survey, so expectations need only be the same on average over the model's estimation period. Moreover, the model may reject the survey information on expectations, if it is not in line with the factors that determine nominal and real yields. So importantly, although our model incorporates survey information, the model forecasts are not always in line with the surveys and may occasionally deviate substantially. And of course compared to the survey information on long-term inflation expectations, which is only available for long horizons every six months, our model has the advantage of providing monthly estimates of expected inflation at any horizon.

3.3. The Kalman filter

We estimate our model by maximum likelihood, using the Kalman filter to compute the log-likelihood function. To apply the Kalman filter, the model needs to be written in state-space form, consisting of a state equation and a measurement equation. We have already described the state equation above, which is the first-order VAR shown in Eq. (15), so we now turn to describe the measurement equation, which shows the relationship between the observed variables and the underlying state variables.

The vector of observed variables consists of month-on-month price inflation, real and nominal bond yields and Consensus forecasts of average inflation 5–10 years ahead. Inflation is relatively straightforward, since we assume it is measured without error. However, we cannot exclude the possibility that government bond yields are measured with error. Measurement error may represent a variety of things, including fitting errors arising from yield curve estimation and market noise. So we include an additional vector of error terms in the measurement equation for each yield to capture such non-fundamental factors. The Consensus survey data we use on long-term inflation expectations are only available every six months (April and October), but this presents no problem in this context because one of the advantages of using the Kalman filter is the possibility of incorporating variables with missing observations into the measurement equation (see, e.g., [Durbin and Koopman \(2001\)](#)). We also include a measurement error on these expectations to allow for the possibility that bond market investors' long-term inflation expectations differ from those of Consensus forecasters.

The measurement equation is therefore given by the following expression:

$$\begin{bmatrix} \pi_t \\ y_t^{i,R} \\ y_t^{i,N} \\ E_t^C(\pi_{t+60:119}) \end{bmatrix} = \begin{bmatrix} \mu_4 \\ \bar{A}_j \\ \bar{A}_i^* \\ \mu_4 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 1 \\ \bar{B}_{j,1} & \bar{B}_{j,2} & \bar{B}_{j,3} & \bar{B}_{j,4} \\ \bar{B}_{i,1}^* & \bar{B}_{i,2}^* & \bar{B}_{i,3}^* & \bar{B}_{i,4}^* \\ G_1 & G_2 & G_3 & G_4 \end{bmatrix} \begin{bmatrix} z_{1,t} - \mu_1 \\ z_{2,t} - \mu_2 \\ z_{3,t} - \mu_3 \\ z_{4,t} - \mu_4 \end{bmatrix} + \begin{bmatrix} 0 \\ u_{t,j} \\ u_{t,i}^* \\ u_{t,c} \end{bmatrix}, \quad (17)$$

where π_t is monthly (seasonally-adjusted) RPI inflation, $y_t^{i,R}$ is the current observed real yield with j months to maturity, $y_t^{i,N}$ is the current observed nominal yield with i months to maturity, $E_t^C(\pi_{t+60:119})$ is the current mean expectation of average inflation 5–10 years

ahead according to Consensus forecasts, $u_{t,j}$ is the measurement error on a real yield with j months to maturity, $u_{t,i}^*$ is the measurement error on a nominal yield with i months to maturity and $u_{t,c}$ is the measurement error on Consensus long-term inflation forecasts. The parameter vectors $\bar{A}, \bar{A}^*, \bar{B}$ and $\bar{B}^*$ embody the theoretical no-arbitrage restrictions, which are functions of $\bar{r}, \gamma, \lambda, \beta, \Omega$ and Φ (derivations are shown in the Appendix). The G_k parameters are determined as $G_k = \frac{1}{60} e_4' \left(\sum_{i=60}^{119} \Phi^i \right) e_k$, where e_k is a (4×1) vector with a 1 in the k th row and zeros in all other rows, so they pick up the factor loadings that predict inflation 5–10 years ahead.

The vector of measurement errors on real yields, nominal yields and Consensus forecasts of inflation are assumed to be independent and normally distributed, with $u_t \sim NID(0, \sigma_R^2 \cdot I_{\#R})$, $u_t^* \sim NID(0, \sigma_N^2 \cdot I_{\#N})$ and $u_{t,c} \sim NID(0, \sigma_c^2)$ respectively.¹² $I_{\#R}$ is an identity matrix of dimension $(\#R \times \#R)$, where $\#R$ is the number of real yields included in the estimation, and $I_{\#N}$ is the corresponding identity matrix with dimension equal to the number of nominal yields included in the model, $\#N$. We assume that the variance of measurement errors is equal across all real yields with different maturities (and equal to σ_R^2) and that the variance of the measurement errors on nominal yields with different maturities is also equal (to σ_N^2). The variance of the measurement errors on Consensus forecasts (equal to σ_c^2) is freely estimated. By having no measurement error in the seasonally adjusted inflation equation, we ensure that the fourth factor is equal to seasonally adjusted inflation. Given the expressions for the state and measurement equations, we can readily apply the Kalman filter to derive the prediction error decomposition of the likelihood function, which can then be maximised over different parameter values to generate maximum likelihood parameter estimates (see, e.g., [De Jong \(2000\)](#)).

4. Data

4.1. Yield curve data

The yield data we use are end-of-month zero-coupon UK nominal and real yields produced by the Bank of England.¹³ The data are calculated using a so-called variable roughness penalty method (see [Anderson and Sleath \(2001\)](#)), which is essentially a cubic spline method with a penalty function that results in the smoothness of the curve increasing with maturity. Nominal yields are based on fitted yields from a curve estimated using general collateral (GC) repo rates and UK nominal government bonds. The real yields data are derived from index-linked and nominal bonds taking into account indexation lags, using the method proposed by [Evans \(1998\)](#) and extended by [Anderson and Sleath \(2001\)](#).¹⁴

Although the Bank of England data for nominal yields are available back to the 1970s and the real yield data back to the mid-1980s, we have restricted our sample period to end-month observations from October 1992 to February 2008 to avoid the potential structural break in term structure behaviour associated with the United Kingdom's adoption of inflation-targeting in October 1992. The lack of short-maturity index-linked bonds over parts of the sample means that real yield data at short maturities tend to be less reliable and are sometimes unavailable, so the shortest-maturity zero-coupon real spot rate we use has a four-year maturity. Thus we model zero-coupon real yields with maturities of 4, 6,

¹² A special case of our model occurs when the inflation expectations of bond market investors are fundamentally different from those of Consensus forecasters, in which case we would expect an estimate of σ_c that is very large.

¹³ The Bank of England publishes UK yield curve estimates on its external website. See <http://www.bankofengland.co.uk/statistics/yieldcurve>.

¹⁴ The method implicitly assumes that there is no indexation lag risk premium on index-linked bonds.

¹¹ [Pennacchi \(1991\)](#) is an early example of the use of surveys, in this case surveys on shorter-term inflation expectations, in a US term structure model.

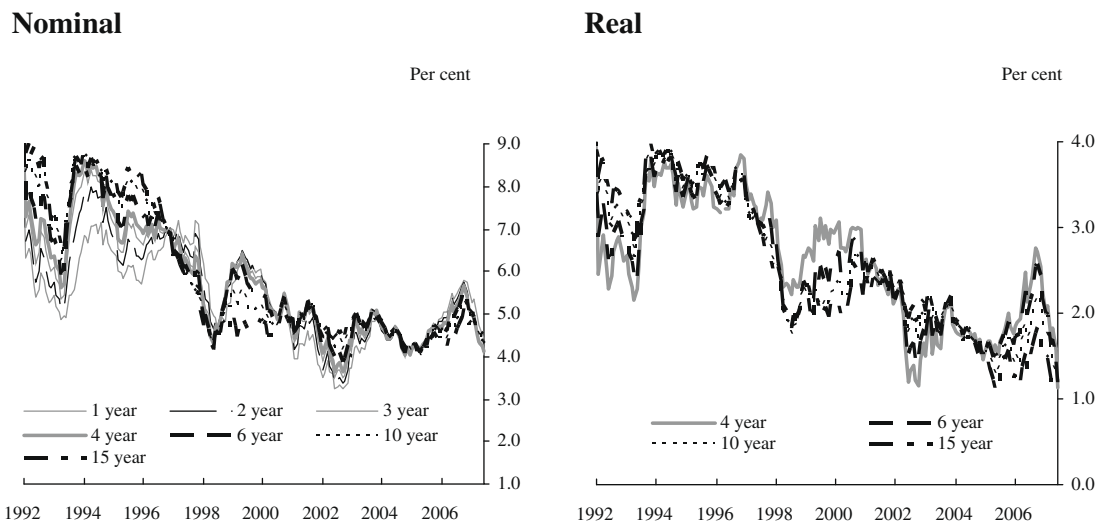


Fig. 1. Nominal and real government (spot) bond yields.

Table 1

Summary statistics for government (spot) bond yields, October 1992 to February 2008.

Maturity (years)	Real yields				Nominal yields						
	4	6	10	15	1	2	3	4	6	10	15
Mean	2.56	2.55	2.53	2.50	5.35	5.51	5.63	5.70	5.78	5.81	5.78
Std Dev	0.70	0.71	0.77	0.85	1.02	1.13	1.22	1.28	1.37	1.49	1.58
Skew	−0.45	−0.05	0.20	0.29	−0.03	0.20	0.39	0.51	0.63	0.73	0.76
Kurtosis	1.94	1.86	1.78	1.74	2.08	2.10	2.20	2.21	2.12	1.93	1.87

10 and 15 years. For nominal yields, we use maturities of 1, 2, 3, 4, 6, 10 and 15 years.

Yields are plotted in Fig. 1. As can be seen, both nominal and real yields have varied significantly over time. Although there is a strong common trend in the yields, there are also periods where short and long-term yields move by different amounts and sometimes in different directions.

Summary statistics for the yields data are shown in Table 1. The UK real yield curve has been flat to very slightly downward sloping on average over the sample, while the nominal curve has been upward sloping. The volatility of real and nominal yields is increasing in maturity. Skewness is slightly increasing in maturity, whereas kurtosis is slightly declining. Principal components analysis, not reported here, of real and nominal yields shows that the first two principal components are able to explain virtually all the variation in real yields with the third factor explaining less than 0.2%. An equivalent principal components analysis of the nominal yields data suggests that three factors are able to explain virtually all the variation in nominal yields, with the fourth factor explaining a mere 0.02%. The main point to take from this analysis is that a relatively few independent factors can explain virtually all the variation observed in nominal and real yields.

4.2. RPI inflation data

As we saw in the theory section above, the link between the nominal and real SDF is the inflation rate with the same frequency as the yield data. As we use monthly data, this means modelling monthly inflation and, as UK real rates are linked to RPI inflation, this means modelling RPI inflation. The strong seasonal pattern in RPI inflation is evident in Fig. 2. But tests reveal that the seasonal pattern is stable and we make a simple seasonal adjustment to the data by running a regression of month-on-month log RPI changes on monthly indicator variables. Investors are assumed to form their inflation expectations using this seasonally adjusted series.

4.3. Survey data

Twice a year Consensus Economics ask their panel of economic and financial forecasters for their forecasts of the underlying rate of UK RPI inflation out to 10 years ahead. The data we include, which refer to forecasts for average expected underlying RPI inflation 5–10 years ahead, are also shown in Fig. 2. The data refer to RPI expectations before 1997 and to RPIX (RPI excluding mortgage interest payments) expectations after 1997. Although there is potentially a problem with the break in the series, RPI and RPIX inflation tend to follow each other at medium to long-term frequencies and any wedge between them in the long-run is likely to be small, probably of the order of 0.1 percentage points. Moreover, as already described in Section 3.3, we allow for the possibility that long-term inflation expectations in the model differ from those of the survey.

As the chart shows, long-term survey expectations have declined over the past 15 years and they have exhibited much less volatility than monthly and annual inflation rates.

5. Results

5.1. Model parameter estimates

As we have four factors and allow all risk prices to depend on each of these factors, the market price of risk has 20 parameters (see Eq. (11)). We started by estimating a completely general model, where all the parameters in the β matrix and λ vector were left unrestricted. We then used a general to specific method, using likelihood ratio tests, to test down until we could no longer reject the joint significance of the parameters.¹⁵ The estimated parameters of our preferred model are shown in Table 2.

¹⁵ Note that where particular parameters do not appear statistically significant according to the reported t -statistics, this implies that the data rejected imposing a zero restriction according to a conventional likelihood ratio test.

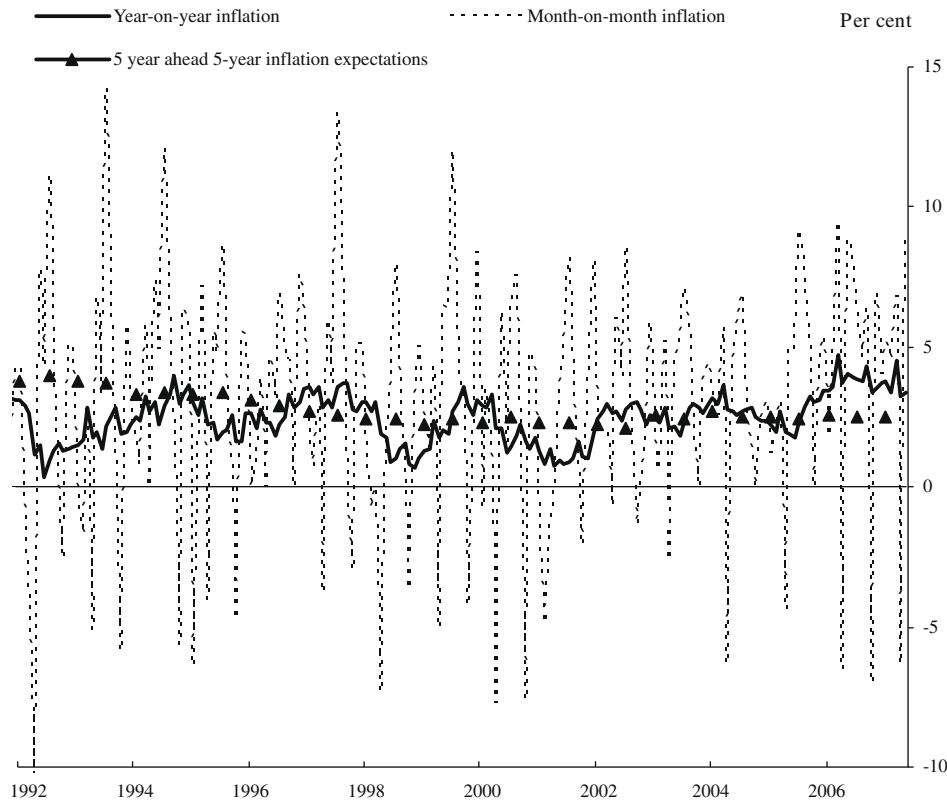


Fig. 2. RPI inflation and Consensus long-term RPI forecasts.

Table 2
Model estimates.

$\begin{bmatrix} \bar{r} \cdot 12 \\ \mu_4 \cdot 12 \end{bmatrix} = \begin{bmatrix} 0.0222 \\ (4.03) \\ 0.0300 \\ (3.45) \end{bmatrix}$	$\Phi = \begin{bmatrix} 0.98 & 0 & 0 & 0 \\ (220.81) & & & \\ 0.002 & 0.99 & 0 & 0 \\ (2.11) & (373.04) & & \\ 0.003 & -0.01 & 0.998 & 0 \\ (1.85) & (1.19) & (1417.87) & \\ 0.28 & -1.71 & -0.22 & 0.11 \\ (3.40) & (2.03) & (1.11) & (1.44) \end{bmatrix}$
$\lambda^* = \begin{bmatrix} -1.180 \\ (0.71) \\ 1.414 \\ (0.98) \\ -0.011 \\ (0.16) \\ -0.083 \\ (1.15) \end{bmatrix}$	$\beta^* = \begin{bmatrix} 0 & 0 & 4800 & 2655 \\ (1.11) & (1.50) & & \\ 0 & 0 & 0 & 9814 \\ & & & (1.63) \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 157 & 0 \\ & & (0.80) & \end{bmatrix}$
$\begin{bmatrix} \sigma_R^* \\ \sigma_N^* \\ \sigma_C^* \end{bmatrix} = \begin{bmatrix} 0.103 \\ (22.63) \\ 0.148 \\ (17.11) \\ 0.195 \\ (6.79) \end{bmatrix}$	$\Omega^* = \begin{bmatrix} 0.15 & 0 & 0 & 0 \\ (12.12) & & & \\ 0 & 0.05 & 0 & 0 \\ & (2.45) & & \\ 0 & 0 & 0.25 & 0 \\ & & (1.11) & \\ 0 & 0 & 0 & 1.88 \\ & & & (15.54) \end{bmatrix}$

t-Statistics in brackets.

 $\lambda^* = \lambda/1000$, $\beta^* = \beta/120$, $\Omega^* = 1000 \cdot \Omega^{1/2}$, $\sigma_k^* = 1200 \cdot \sigma_k$ for $k = N, R, C$.

Given the reduced-form nature of the model, it is difficult to give a meaningful structural interpretation to most of the estimated parameters. The key point is that risk premia are found to display significant time-variation, as can be seen from the estimated parameters in the β matrix. From the parameter estimates it seems evident that the first, second and fourth risk prices change over time and, in fact, we cannot reject a joint test of time-variation in these risk prices. Interestingly, we find that it is the two additional nominal factors that determine all the time-variation in the market price of risk.¹⁶ Consistent with other studies, the la-

tent factors are all quite persistent – the persistence of the third factor being particularly strong. From the estimated Φ matrix, it is also noteworthy that inflation is forecastable in-sample. Perhaps surprisingly, it is not the lag of inflation which is strongly significant, but the lag of the first and second factors. Finally, it is worth mentioning that the variance of the measurement error of the Consensus survey data on inflation expectations is estimated relatively precisely, so we cannot reject the inclusion of the survey data. We will shed further light on the estimated parameters below.¹⁷

5.2. Model forward rate decompositions

Fig. 3 shows the model-implied decompositions of the instantaneous forward rate, 4 years and 10 years ahead, and the component that is left unexplained by the model.¹⁸ The nominal forward rate decompositions in the first panel show that the model fits nominal forwards well, with the residual lines quite close to zero for most of the sample period. This is particularly impressive, as the model was optimised to fit spot interest rates rather than forward rates per se, so in this sense these forward-rate residuals are out of sample. Over the sample as a whole, nominal forward rates at medium and long-term horizons have fallen, with most of the fall occurring in the period from 1997 to 1998. The model mainly attributes this fall to lower term premia, although longer-horizon expected short nominal interest rates have also fallen over the period.

¹⁷ To judge the performance of the model, we examined a number of summary statistics of the in-sample fit of the model. These suggest the model fits the data reasonably well, with an average measurement error close to zero and a relatively low standard deviation. While there is some serial autocorrelation in the measurement errors, it is broadly comparable to similar models (see, e.g., De Jong (2000)).

¹⁸ By instantaneous rate, we mean the one-month rate, so the 10-year instantaneous forward rate is the one-month rate implicit 10 years in the future. Note that we include convexity effects in our term and inflation risk premia estimates, as the latter are constant over time and small (e.g., less than 12 basis points in absolute terms for the 15-year nominal forward rate).

¹⁶ Balfoussia and Wickens (2007) also find that inflation strongly outperforms any real variables in explaining the SDF in an empirical application to US data. But as they use a slightly different set-up, with time-varying covariances between bond returns and inflation, it is difficult to make an exact comparison.

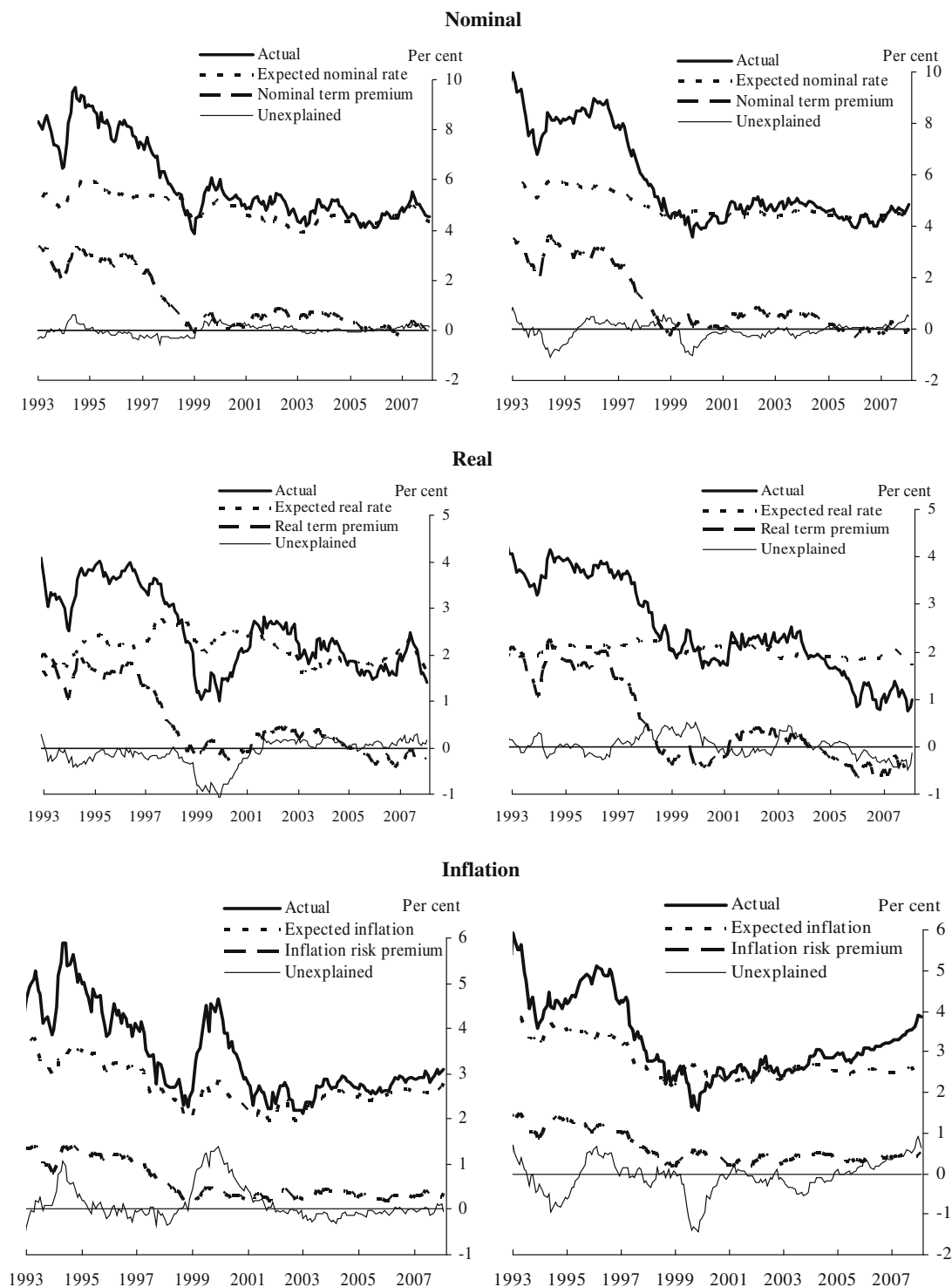


Fig. 3. Decomposition of 4-year and 10-year forward rates.

If we look at the second and third panels of the chart, we see the corresponding decompositions for real and inflation forwards. Again the model fits these forward rates well, although there are a few periods where there is a more pronounced deterioration in the model's ability to fit the data (see discussion below). Medium to long-term inflation expectations have declined over the sample and have remained at relatively low and stable levels since 1997–98. Long-horizon expected real short rates have varied less over the sample, with an average level of 2.1% (about 2.9% after adjusting

for an estimate of the long-run wedge between RPI and CPI inflation) with the lowest value being 1.7% and the highest value around 2.3%. It is apparent that the fall in nominal premia up to 1997–98 shown in the first panel is accounted for by both lower real term premia and lower inflation risk premia, although it is falls in the former that have dominated the decline in nominal premia. Most of the variation since 1998 in nominal term premia is also attributed to changing real term premia, while inflation risk premia have been quite stable.

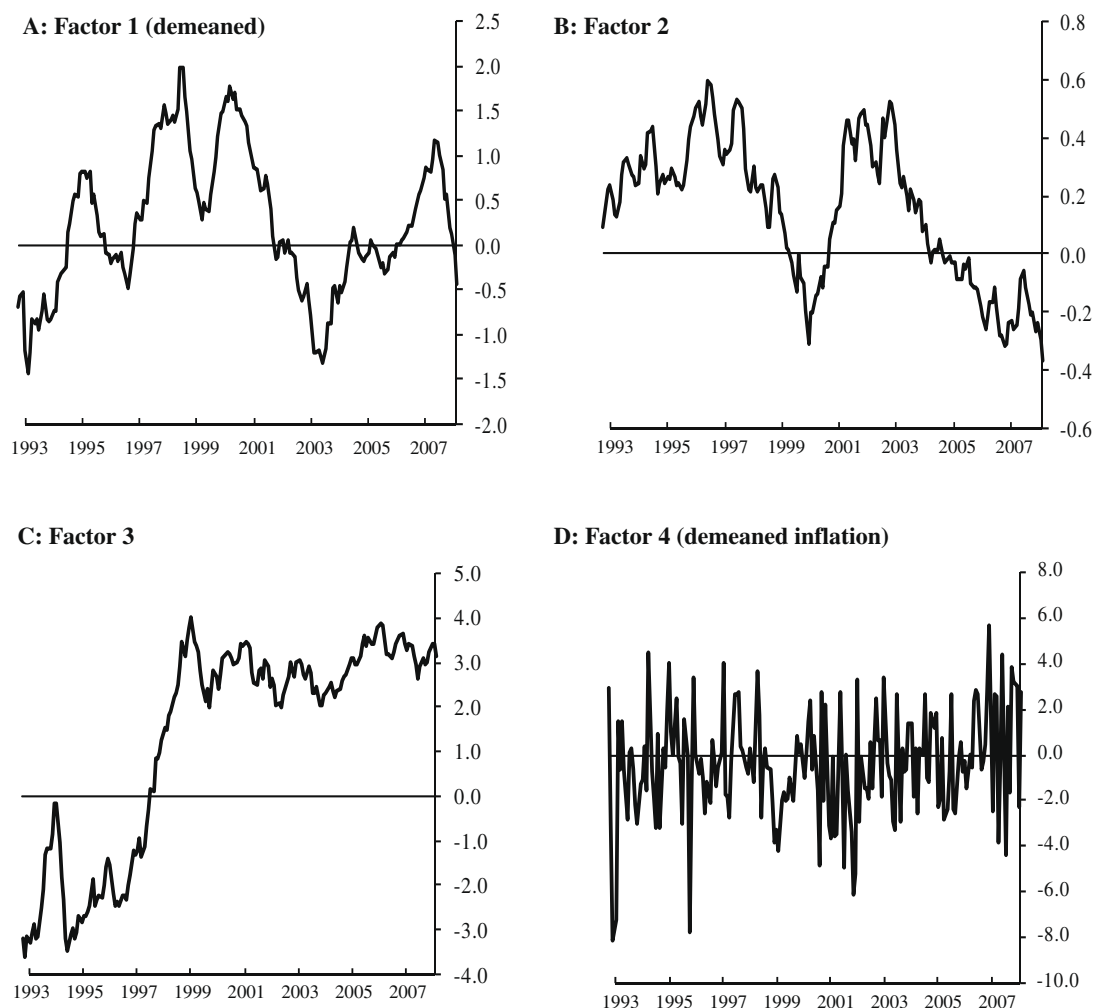


Fig. 4. Factors driving the yield curve (multiplied by 1200).

We note that although the model fits well over most of the sample, it tends to do less well in periods that coincide with severe turbulence in financial markets such as the Asian and LTCM crises in 1997 and 1998, the troubles of IT stocks in the early 2000s and the more recent market disruptions since the middle of 2007, originally triggered by the sub-prime mortgage crisis in the United States. One potential explanation for this could be the impact of large ‘flight to quality’ flows into government bonds during these periods, which may have led to temporary changes in liquidity premia, which are not explicitly modelled in our framework.¹⁹

The factors themselves are shown in Fig. 4. Panels A, B and C reveal that, although the three latent factors are highly persistent, their time-series dynamics are rather different. More revealing perhaps are the impulse responses shown in Fig. 5, which enable us to analyse the impact on forward rates of a one standard deviation shock to each of the factors. The loadings suggest that the first factor affects the slope of both the real and nominal forward curves, with its impact coming through expected real short rates and inflation. The second factor primarily affects the level of the real forward curve, with the effect coming through expected real

short rates (declining with horizon) and real term premia (where the effect increases with horizon), although it also impacts on nominal forward rates mainly through expected inflation. The third factor affects the curvature of both the real and nominal forward curves, with its impact coming entirely through expected inflation and inflation risk premia and real term premia. Combining the results from these figures and Fig. 3 suggests that this factor helps explain the fall in expected inflation, real term premia and inflation risk premia in 1997–1998. The impulse responses also reveal that RPI inflation shocks have their strongest positive impact on very short-term forward rates, through their impact on short-term inflation expectations. Interestingly, inflation shocks do not impact on medium or long-term real and nominal forward rates, which is consistent with what we would expect to see under a credible monetary policy framework. As was apparent from the estimated parameters of the model, inflation shocks push down on the short-term real term premium, perhaps indicating increasing demand for real indexed-linked bonds in periods of relatively large inflation shocks. But importantly the impulse responses show that this fall in real term premia is relatively small and concentrated at horizons out to 24 months.

To quantify more precisely the main drivers of nominal forwards, real forwards and implied inflation breakevens at different horizons, Table 3 contains a simple variance decomposition breakdown over the full sample period. From the results in the first

¹⁹ It could also reflect limitations arising from the linear nature of the model. It seems plausible that the importance of nonlinearities in explaining term premia (as suggested by the literature on the importance of a jump risk; see, e.g., Wright and Zhou (2009)) may be more important in situations of extreme market volatility.

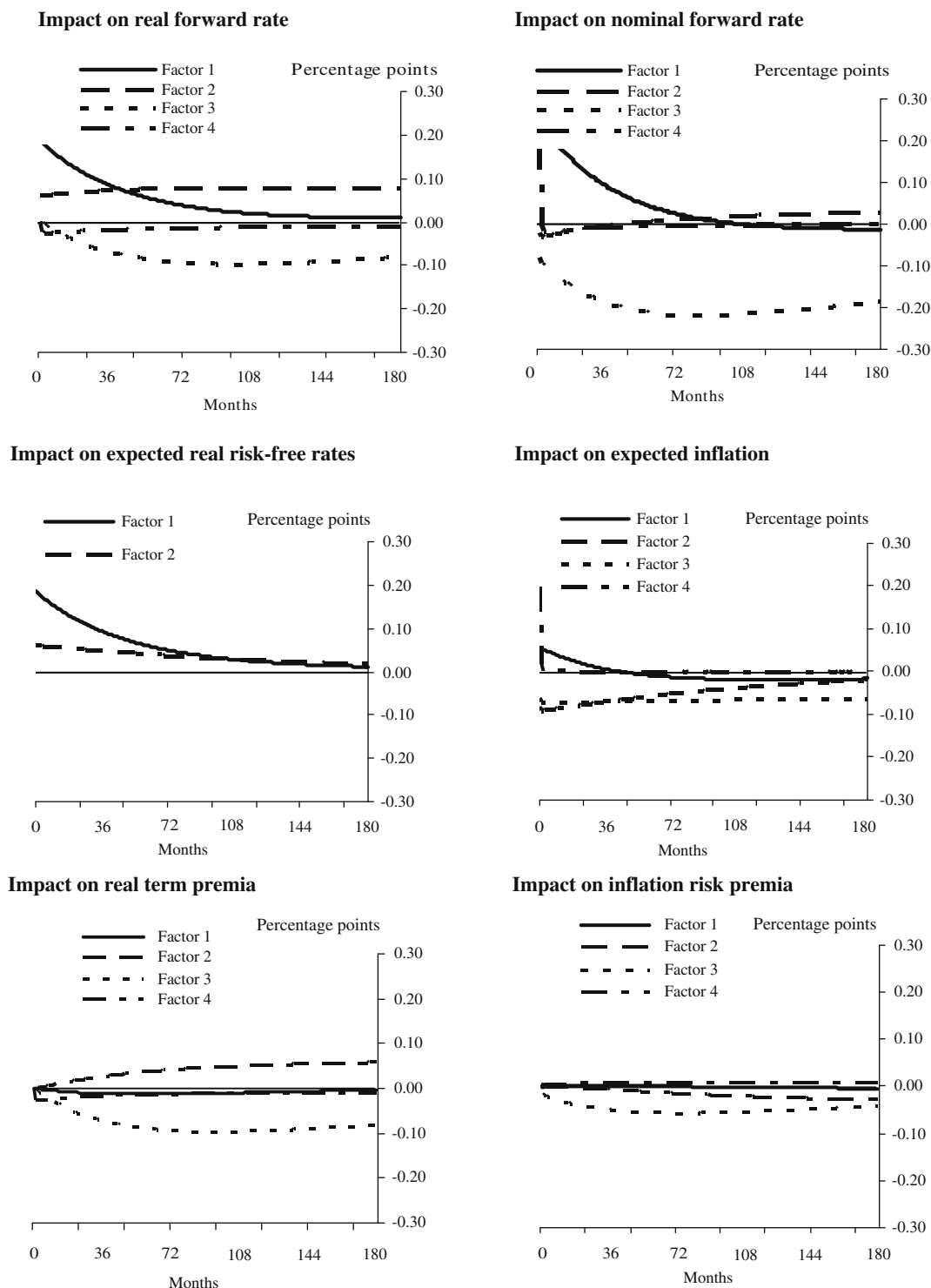


Fig. 5. Factor loadings on forward rates.

panel, we can see that expected real short rates and expected inflation account for about 60% of the variance of nominal 1-year forward rates. While the contribution of expected real short rates is broadly equal to the contribution of expected inflation at 1-year horizons, it tails off at medium to long horizons. In contrast, expected inflation accounts for a similar amount of the variance (about 25%) in both 4-year and 10-year nominal forward rates. Real term and inflation risk premia explain more than half of the variance in medium to long-term nominal forward rates, with real

term premia accounting for roughly twice as much as inflation risk premia. In terms of monthly movements (Panel 2), the model accounts for about three quarters of the variance of nominal forward rates.

Turning to the decomposition for real forward rates (Panel 3), we first need to note that the decomposition of the variance of 1-year real rates has to be based on model-implied rates, as we do not have data for 1-year real (or inflation) forwards. Nevertheless, it is interesting that real term premia account for as much as

Table 3
Variance decompositions (%).

$f_t^{n,N}$	$E_t(y_{t+n}^{1,R})$	$E_t(\pi_{t+n})$	$\varphi_{r,t}^n$	$\varphi_{\pi,t}^n$	$u_t^{n,N}$
$n = 12$	28.40	30.40	16.00	17.40	7.80
$n = 48$	3.70	25.10	45.20	26.60	−0.60
$n = 120$	1.70	24.90	48.50	22.30	2.60
$\Delta f_t^{n,N}$	$\Delta E_t(y_{t+n}^{1,R})$	$\Delta E_t(\pi_{t+n})$	$\Delta \varphi_{r,t}^n$	$\Delta \varphi_{\pi,t}^n$	$\Delta u_t^{n,N}$
$n = 12$	32.40	27.50	5.80	7.30	27.00
$n = 48$	7.40	20.40	28.10	18.50	25.60
$n = 120$	0.70	20.50	37.00	17.90	23.90
$f_t^{n,R}$	$E_t(y_{t+n}^{1,R})$		$\varphi_{r,t}^n$		$u_t^{n,R}$
$n = 12$	82.80		17.20		
$n = 48$	10.00		81.70		8.30
$n = 120$	5.20		91.00		3.80
$\Delta f_t^{n,R}$	$\Delta E_t(y_{t+n}^{1,R})$		$\Delta \varphi_{r,t}^n$		$\Delta u_t^{n,R}$
$n = 12$	89.40		10.60		
$n = 48$	29.90		36.10		34.00
$n = 120$	8.80		64.80		26.40
BE_t^n	$E_t(\pi_{t+n})$		$\varphi_{\pi,t}^n$		$u_t^{n,BE}$
$n = 12$	66.40		33.60		
$n = 48$	43.00		39.20		17.80
$n = 120$	44.10		28.30		27.60
ΔBE_t^n	$\Delta E_t(\pi_{t+n})$		$\Delta \varphi_{\pi,t}^n$		$\Delta u_t^{n,BE}$
$n = 12$	81.00		19.00		
$n = 48$	31.50		19.90		48.60
$n = 120$	22.60		17.50		59.90

$\varphi_{r,t}^n$ denotes the real term premium; $\varphi_{\pi,t}^n$ denotes the inflation risk premium.

17% of the variance, suggesting that we need to adjust for time-varying real term premia even at quite short horizons if we wish to measure real policy rate expectations. At medium and long-term horizons, expected real short rates make little contribution and real term premia dominate. The decomposition of breakevens (Panel 5) suggests that as much as a third of the variance of 1-year rates is attributable to inflation risk premia (although again note that this is based on model-implied rates, so there is no residual category). At medium and longer-term horizons inflation premia remain important, but over 40% of the variance is attributed to inflation expectations. In terms of the variance of month-to-month movements (Panels 4 and 6), we find that the model is more successful in explaining real forward rates than in explaining inflation forward rates.

The term structure decompositions in Fig. 3 suggest that there are two periods in particular that merit more analysis. The first is the 1997–98 period, following the granting of operational independence to the Bank of England in setting interest rates. The second is the period since 2004, which accompanied the emergence of the bond market conundrum, where long-horizon real rates fell to historically low levels. From the model decompositions shown in the second panel of Fig. 3, it is clear that this accompanied both a fall in real term premia (to negative levels) and an increase in the unexplained component of long-horizon real rates, as the model over-predicted real forward rates. We discuss these issues further below.

5.2.1. The impact of Bank of England independence

We have already shown that forward rates fell sharply around the time that the Bank of England was granted operational independence for setting interest rates. Fig. 6A and B shows the forward-rate decompositions from the model out to 15 years, if we average over the years in the sample before and after Bank independence. Fig. 6A shows that average expected inflation is lower

across all maturities post independence. Fig. 6B shows that the average level of the forward curve of inflation risk premia is lower as well, with the fall at 10-year horizons of the order of 70 basis points. In contrast, while average expected short to medium-term real rates have been higher since Bank independence, mainly due to the higher level of expected real short rates between 1997 and 2001, little has happened to the level of expected long-horizon real short rates. But the average level of real term premia has fallen significantly and has, on average, been negative at longer horizons since Bank independence.

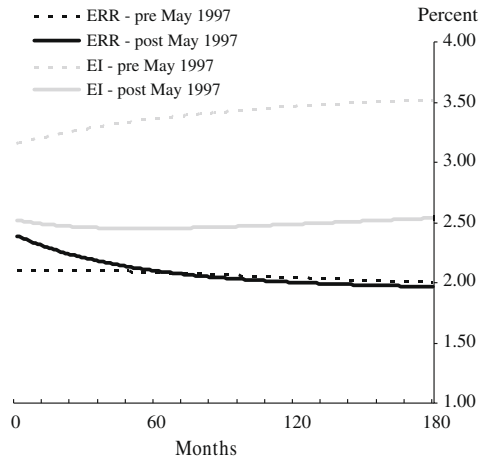
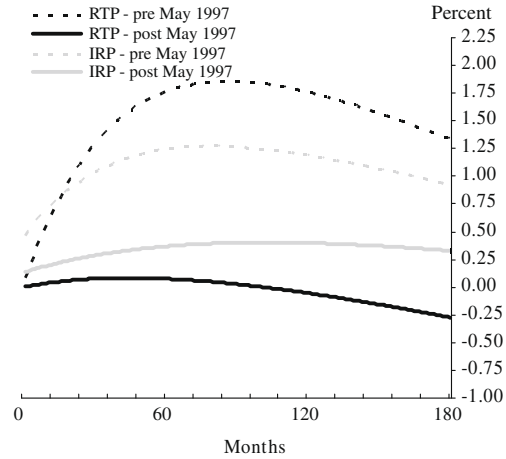
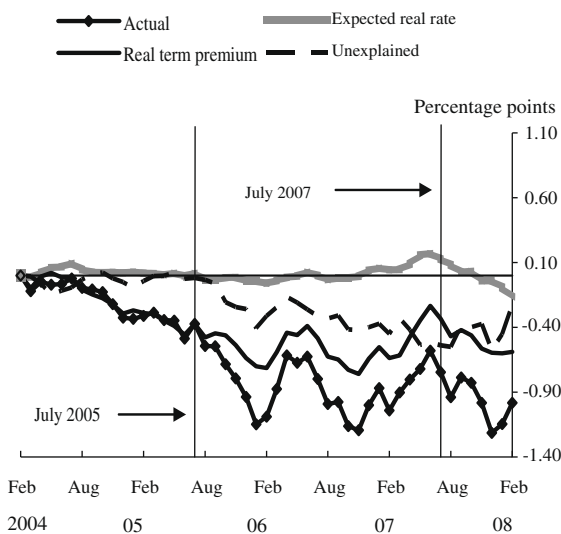
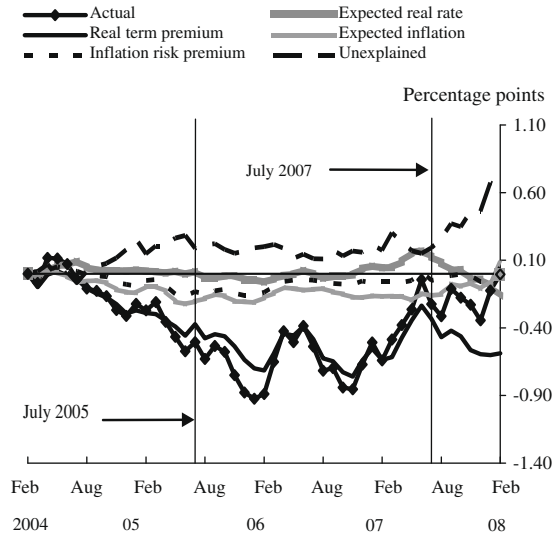
The large fall in the level of real term premia is unlikely to have been mainly related to the independence of the Bank and could reflect a number of other events that affected bond markets over this period. One important contributory factor is likely to have been the introduction of the Minimum Funding Requirement, part of the 1995 Pensions Act, which became effective in April 1997. This regulatory reform, designed to protect the solvency of pension funds, led to increased pension fund demand for index-linked bonds, which seems to have compressed measured real term premia. And this may have been reinforced by the Asian and LTCM crises in 1997 and 1998, which caused a ‘flight to quality’ into government bonds. At around the same time, there was also an expansion of the global index-linked bond market, with the first issuance of US Treasury Inflation Protected Securities (TIPS) in January 1997 (see Elsasser and Sack (2004)), which may have reduced the liquidity premium attached to index-linked bonds as an asset class. Perhaps unsurprisingly, our model has more difficulty in explaining real forward rates over this period.

An important question is whether the changes surrounding independence constitute a structural break in the data. This is difficult to test formally, as the sample over which the model is estimated is quite short. Also since our model is not structural and contains latent factors, which we might expect to pick up any structural changes, it is hard to assess the extent to which the various relationships (e.g. the relationship between inflation expectations, inflation risk premia and the factors) have changed. But, from the shocks to the three latent factors and actual RPI inflation (not shown), there is little to suggest that there has been a structural change in the relationship between the factors explaining yields.

5.2.2. The bond yield conundrum

The more recent bond market conundrum refers to the large fall in international medium to long-term yields which began in the middle of 2004. Joyce et al. (2008) estimate an essentially affine real term structure model, which allows a decomposition of the UK real yield curve into expected future real short rates and real term premia. Based on a number of different specifications, they conclude that the main reason for the large fall in long-term real forward rates over this period was a fall in real term premia. Our model of the nominal and real yield curves adds additional structure to their analysis and allows us to reinterpret the reasons for the large fall in long-term real forward rates.

In Fig. 7 we plot cumulative changes in the model-implied components of 10-year nominal and real forward rates since early 2004. It is useful to distinguish between three different periods: before July 2005, between July 2005 and July 2007 and after July 2007. Consistent with the findings of Joyce et al. (2008), we find that a large fall in the real term premium accounts for most of the fall in the 10-year real forward rate up to July 2005. But, in contrast to their results, we find that our model cannot account for much of the fall in the long-term real forward rate between July 2005 and July 2007. Over this period, the more negative contribution from the unexplained part of the model indicates that the actual 10-year real forward rate became lower than the model would predict. But over the same period there is no deterioration in the model’s ability to fit the nominal rate. Our model decomposition

A: Expected real risk-free rates (ERR) and expected inflation (EI)**B: Inflation risk premia (IRP) and real term premia (RTP)****Fig. 6.** Forward curves of expectations and real term/inflation risk premia.**A: Real****B: Nominal****Fig. 7.** Cumulative changes in 10-year forward rates.

therefore suggests that there were factors that were pushing down on the expected real short rates or real term premia embodied in the real yields, which did not affect the expected real short rates or real term premia embodied in nominal forward rates. Obviously this could be a symptom of model misspecification, perhaps related to the fact we have assumed that bond markets are not segmented at different maturities. There are indeed reasons to believe that the longer end of the index-linked bond market may have become more segmented during this period, as a result of a number of regulatory developments.²⁰ By encouraging pension funds to

match their long-term liabilities, these changes may have led to an increase in institutional demand for long-maturity index-linked bonds (see McGrath and Windle (2006)). So on this interpretation, the large negative residual on long-term real forwards might reflect the fact that institutional investors were constrained to buy and hold longer-term index-linked bonds for regulatory reasons, even if their price exceeded the price consistent with fundamentals at the time.

After the start of the financial market turbulence in July 2007, however, the story changes somewhat. While the unexplained part of the real forward rate becomes smaller in absolute terms, the unexplained part of the nominal rate becomes larger. In other words, while actual long-term real forward rates are more in line with the model's predictions, the opposite is true for nominal bonds. But one should take care with drawing strong conclusions based on the model results since July 2007. Our model is better suited to capturing longer-term trends in the data and, as we noted above, has particular problems in fitting bond yields during periods of severe financial market turbulence.

²⁰ At the beginning of 2005 the FRS17 'Retirement benefits' standard became mandatory in the United Kingdom, requiring pensions schemes to value their assets at market value and for scheme surpluses/deficits to be recognized explicitly on company balance sheets. In April of the same year the Pension Act 2004 came into force and the Pension Protection Fund (PPF) and pensions regulator were launched. A risk-based levy consultation document was subsequently published by the PPF in July 2005, but perhaps more significantly new funding regulations came into force at the end of 2005, replacing the requirements under the Minimum Funding Requirement (referred to earlier), and pension fund trustees were given up to 18 months to complete valuations of their funds on this new basis (see Bank of England (2006)).

Concluding on our model-based reasons for the fall in real rates, we find that, of the 100 basis points or so fall in the 10-year real forward rate over the period since the beginning of 2004, around 60 basis points was accounted for by a lower real term premium and about 20 basis points by a lower expected real short rate, leaving 20 basis points unexplained. So the long-term real rate remained at a slightly lower level than the model would predict at the end of our sample period. The inability of the model to explain the large fall in long-term real forward rates during July 2005 to July 2007 is also mirrored in its inability to account for the increase in implied long-term forward inflation rates over the same period, perhaps suggesting that special factors in the index-linked market might partly explain both developments. Although our model-implied long-term inflation expectations did pick up slightly in late 2007 to early 2008, during July 2005 to July 2007 both inflation expectations and inflation risk premia were broadly stable.

Obviously we need to be careful when making structural interpretations of the unexplained part of the model and the reasons for changes in the various components of forward rates. To give a more meaningful interpretation of the dynamics of interest rates, we would need a more formal general equilibrium model. But, to address the issue of potential segmentation of the long end of the real yield curve we have described, such a model would also need to incorporate heterogeneous agents. Needless to say such models are not yet well developed (for an attempt to incorporate bond market heterogeneity into a partial equilibrium model, see, e.g., Vayanos and Vila (2007)).

6. Conclusions

In this paper, we develop a joint, essentially affine, model of the UK real and nominal term structures, which allows us to decompose forward rates into expected real short rates, expected inflation, real term premia and inflation risk premia. To our knowledge, this is the first study to estimate an essentially affine no-arbitrage model of this nature for the United Kingdom over the period since October 1992, when the UK monetary authorities adopted an explicit inflation target.

The model set-up implies that the market price of risk is linear in a small set of latent factors and observable RPI inflation. The advantage of using latent factors is the ability of this approach to fit nominal and real yields well, at both long and short horizons. However, due to the use of latent factors and small sample problems, we also argue that it is important to include survey measures of long-term inflation expectations as additional information in the model, in order to improve the identification of the dynamics of long-term expectations and term premia. Importantly, since we allow long-term inflation expectations to equal survey expectations with an error, we also effectively allow the term structure model to reject the surveys, if they are inconsistent with the information implicit in the term structure and the factors driving nominal and real yields. And, by including a large number of yields, we do not force the model to give a disproportionate weight to fitting the surveys.

An advantage of the joint model is that it can be used to decompose nominal rates into their various real and nominal components using market data on both conventional and index-linked government bonds. Based on our analysis, we find that expected real short interest rates and expected inflation together explain around 60% of the variation in 1-year nominal forward rates, but at longer horizons inflation risk premia and, in particular, real term premia explain a much larger fraction of the variation.

We use the model to analyse the impact of Bank of England independence on the term structure of interest rates and the large fall in UK long-term real interest rates since 2004. We find that both longer-term inflation risk premia and inflation expectations

embodied in the term structure have been lower since the Bank of England was granted operational independence for setting interest rates in May 1997. More recently, we find that the conundrum of unusually low long-term real rates can mainly be attributed to a fall in real term premia, although a significant part of the fall in rates is left unexplained, particularly over the period between July 2005 and July 2007. We suggest that the relative inability of the model to fit long horizon real forwards during this period might partly be a consequence of strong pension fund demand for index-linked bonds arising from recent regulatory changes. Moreover, the model decompositions suggest that these special factors affecting the index-linked market might also help explain the rise in longer-horizon inflation breakeven rates that occurred at around the same time. One challenge for future research will be to incorporate financial market frictions and preferred habitat behaviour more formally into empirical models of the term structure.

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Appendix

A.1. Real yields and no arbitrage

Starting from the fundamental asset pricing equation, the price of a real bond with n periods to maturity is given by

$$p_t^{n,R} = E_t \left[M_{t+1} P_{t+1}^{n-1,R} \right]. \quad (A1)$$

Taking logarithms of both sides of this equation:

$$p_t^{n,R} = E_t \left[m_{t+1} + p_{t+1}^{n-1,R} \right] + \frac{1}{2} \text{Var}_t \left[m_{t+1} + p_{t+1}^{n-1,R} \right], \quad (A2)$$

where lower-case letters denote logs. With the assumed real SDF (Eq. (9)), the real bond price will be given by

$$p_t^{n,R} = A_n + B'_n (z_t - \mu),$$

where $B_n = [B_{n,1} \ B_{n,2} \ B_{n,3} \ B_{n,4}]$ is a (4×1) vector, $(z_t - \mu)$ is a (4×1) vector and A_n is a scalar. If we now substitute in for next period's SDF and for the bond price, it is possible after some algebraic manipulation to get to this expression, where A_n and B'_n are defined recursively by

$$A_n = -\bar{r} + A_{n-1} - B'_{n-1} \Omega \lambda + \frac{B'_{n-1} \Omega B_{n-1}}{2}, \quad (A3)$$

$$B'_n = -\gamma' + B'_{n-1} (\Phi - \Omega \beta), \quad (A4)$$

where λ and β are defined in Eq. (11) and are matrices of dimensions (4×1) and (4×4) , respectively. The price of a bond maturing today is $p_t^{0,R} = 1$, so the initial condition is simply:

$$A_0 = 0 \quad B'_0 = [0 \ 0 \ 0 \ 0]. \quad (A5)$$

This yields the recursive relationship between yields of different maturities obeying the no-arbitrage restriction. Real yields, continuously compounded, are given by

$$y_t^{n,R} = -\frac{p_t^{n,R}}{n} = -\frac{A_n}{n} - \frac{B'_n}{n} (z_t - \mu) = \bar{A}_n + \bar{B}'_n (z_t - \mu). \quad (A6)$$

The one-period real rate, for instance, is therefore given by

$$y_t^{1,R} = \bar{r} + \gamma'(z_t - \mu). \quad (\text{A7})$$

A.2. Nominal yields and no arbitrage

The relation between the price of a nominal bond today with n periods to maturity and the price of this bond in the next period when it has $n - 1$ periods to maturity is given by

$$P_t^{n,N} = E_t \left[M_{t+1} P_{t+1}^{n-1,N} \frac{Q_t}{Q_{t+1}} \right]. \quad (\text{A8})$$

Taking logarithms of both sides of this equation we get

$$p_t^{n,N} = E_t \left[m_{t+1} + p_{t+1}^{n-1,N} - \pi_{t+1} \right] + 1/2 \text{Var}_t \left[m_{t+1} + p_{t+1}^{n-1,N} - \pi_{t+1} \right], \quad (\text{A9})$$

where lower-case letters denote logs. Using $\pi_{t+1} = E_t[z_{4,t+1}] + \sigma_4 \varepsilon_{4,t+1}$ and $E_t[z_{4,t+1}] = E_t[\pi_{t+1}] = \mu_4 + \Phi_{[4,:]}(z_t - \mu)$, the nominal SDF is given by

$$m_{t+1}^* = -(\bar{r} + \mu_4) - (\gamma' + \Phi_{[4,:]})(z_t - \mu) - \frac{\Lambda_t' \Omega \Delta_t}{2} - \Lambda_t' \Omega^{1/2} \varepsilon_{t+1},$$

where $\Phi_{[4,:]}$ refers to the fourth row of Φ and

$$\Lambda_t^* = \lambda^* + \beta z_t = \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 + 1 \end{bmatrix} + \begin{bmatrix} \beta_{11} & \beta_{12} & \beta_{13} & \beta_{14} \\ \beta_{21} & \beta_{22} & \beta_{23} & \beta_{24} \\ \beta_{31} & \beta_{32} & \beta_{33} & \beta_{34} \\ \beta_{41} & \beta_{42} & \beta_{43} & \beta_{44} \end{bmatrix} (z_t - \mu)$$

The log of nominal bond prices is linear in the factors:

$$p_t^{n,N} = A_n^* + B_n^{*'}(z_t - \mu), \quad (\text{A10})$$

where

$$A_n^* = -\bar{r} - \mu_4 + A_{n-1}^* - B_{n-1}^{*'} \Omega \lambda^* + \frac{B_{n-1}^{*'} \Omega B_{n-1}^*}{2} + \frac{\sigma_4^2}{2} + \sigma_4^2 \lambda_4 \quad (\text{A11})$$

and

$$B_n^{*'} = -(\gamma' + \Phi_{[4,:]}) + B_{n-1}^{*'}(\Phi - \Omega\beta) + e_4' \Omega \beta. \quad (\text{A12})$$

Using $P_t^{0,N} = 1$, i.e. the price of a nominal bond maturing today has a price equal to one, the initial condition is given by

$$A_0^* = 0 \quad B_0^{*'} = [0 \quad 0 \quad 0 \quad 0]. \quad (\text{A13})$$

Nominal yields, continuously compounded, are therefore linear in the demeaned factors and given by

$$y_t^{n,N} = -\frac{p_t^{n,N}}{n} = -\frac{A_n^*}{n} - \frac{B_n^{*'}}{n}(z_t - \mu) = \bar{A}_n^* + \bar{B}_n^{*'}(z_t - \mu). \quad (\text{A14})$$

The one-period nominal rate is given by

$$y_t^{1,N} = y_t^{1,R} + \mu_4 + \Phi_{[4,:]}(z_t - \mu) - \frac{\sigma_4^2}{2} - \sigma_4^2 \lambda_4 + e_4' \Omega \beta (z_t - \mu). \quad (\text{A15})$$

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